

Practice Problems

Practice problems: NPV

- **SolarCo considers** investing in a solar panel production machine, which costs \$750k and generates \$150k in annual cash flows forever (in perpetuity). SolarCo has already spent \$80k on a feasibility study. Moreover, if they get the machine, they must give up \$25k per year in warehouse rent from the space the machine occupies.
 - What is the NPV of investing in the solar panel production machine?
- **SolarCo is considering** setting up a research lab to develop its own solar panel machine. The lab costs \$800k to set up. It has a 50-50% chance of successfully developing a machine. If it succeeds, the machine will increase annual cash flows by \$200k, on average, forever (in perpetuity)
 - What is the NPV of investing in the research lab?
- **SolarCo now faces a choice** between two machines. Machine A costs \$600k and improves annual cash flow by \$100k. Machine B costs \$900k and improves annual cash flow by \$120k. Which one of the two machines should SolarCo choose?

Assumptions: Appropriate risk-adjusted discount rate is 10%

What is a Company?

Dutch East India Company (1602)

- **World's first modern company was the Dutch East India Company**, founded in the Netherlands in 1602 to trade in the Indian Ocean and assist in the Dutch war of independence against Spain
- Company prospered through most of the 17th century as the Dutch empire in the East Indies grew. It was dissolved in 1799
- Company's operations were so large and profitable that some historians estimate it was the most valuable company ever in history. Inflation adjusted, its value was over \$7 trillion at its peak



Dutch East India Company (1602)

It was first company with modern features:

- **Joint-stock ownership:** Shares were freely tradable on the Amsterdam Stock Exchange, which was also the first of its kind. *it cannot go negative*
- **Limited liability:** Shareholders were not personally liable for the company's debts.
- **Permanent capital:** Unlike earlier ventures, which dissolved after a voyage or a few years, it had a permanent structure.



Modern C-corporations (C-corps)

- US publicly-traded companies are typically C-corps
- C-corps are owned by their shareholders
 - Ownership is divided into individual shares of stock
 - Shares can be freely traded, if the company is publicly listed
 - Shareholders have limited liability
- Governance Structure
 - Company is controlled by a board of directors, elected by the shareholders
 - Shareholders have one vote for each share
 - Board appoints CEO to run company
 - Board and CEO have “fiduciary duty” to act in shareholders’ interest
- Unlimited Life
 - Infinitely lived, in principle



Corporate governance

- **Shareholder view:** Goal of a company is to maximize value to shareholders
 - Also known as “Friedman doctrine” (“the business of business is business”): Executives are agents of shareholders and should act in their interests, not pursue personal or social goals. If shareholders would want to pursue such goals, they can do so on their own, with their own money
- **Stakeholder view:** Companies should serve stakeholders broadly, including employees, customers, suppliers, the environment, etc.
 - In 2019, US Business Roundtable endorsed the stakeholder view: “Each of our stakeholders is essential. We commit to deliver value to all of them, for the future success of our companies, our communities, and our country.”
- In many situations these two views are not in contradiction

Other company types

- C-Corporation (C-corp.) (“normal” company)
 - Owners are the shareholders with limited liability
 - Pays corporate taxes (double taxation)
- Sole Proprietorship (around 70% of all businesses)
 - Owned and operated by single individual
 - Simple to set up and operate
 - No separation between individual and company
 - Owner has unlimited liability
 - Tax pass-through to income taxes
- Cooperative (Co-op)
 - Owned and controlled by members
 - One member, one vote (not one share, one vote)
 - Profits are distributed among members
- Non-profit Corporation
 - Educational, religious, charitable, or public organizations
 - Largely tax exempt
- Other company types: LLCs, B-corps, S-corps, partnerships (and there are a few more)

DCF/NPV in Practice: Free Cash Flows (FCFs)

Free Cash Flows (FCFs)

- When valuing a project (or a company) we use the “Free Cash Flows” (FCF)
 - Also sometimes called “Unlevered” or “All-equity” Free Cash Flows
- FCFs are cash flows that are:
 - “Free” to be paid out, if needed
 - Actual cash flows (not accounting earnings)
 - Incremental (account for opportunity costs and sunk costs)
 - Expected cash flows (statistical average)
 - Exclude interest payments (“Unlevered” or “All-equity”)

Projecting (forecasting) FCFs

- **Forecast** future financials in an income statement
 - **Income statement projects future:** Revenue, Cost of Goods Sold (COGS), Sales, General & Administrative expenses (SGA), Capital Expenditures (CAPX), Depreciation (DEPR), Taxes (TAX) and Net Working Capital (NWC)
- **Then**, calculate Free Cash Flows (FCFs) from the income statement figures

What are the Free Cash Flows (FCFs)?

The 1999 income statement of Executive Paper Corporation (figures in \$ millions)

	\$ MILLIONS
Revenues <i>cash flow</i>	2,200
Costs	1,980
Depreciation	53.3
EBIT	166.7
Interest	40
Tax	50.7
Net income	76.0
Dividends	45.6
Retained earnings <i>cash flow</i>	30.4
Earnings per share, dollars	5.37
Dividend per share, dollars	3.22

Example: Income Statement

Item (figures in \$'000)		Year 0	Year 1	Year 2	Year 3
Revenue (REV)		\$0.0	\$400.0	\$440.0	\$480.0
Cost of Goods Sold (COGS)		\$0.0	\$220.0	\$240.0	\$260.0
Gross Profit (REV-COGS)		\$0.0	\$180.0	\$200.0	\$220.0
Operating Expenses (SG&A)		\$0.0	\$120.0	\$120.0	\$120.0
Depreciation (DEPR)		\$0.0	\$40.0	\$40.0	\$40.0
EBIT (Gross Profit-SG&A-DEPR)		\$0.0	\$20.0	\$40.0	\$60.0
Capital Expenditure (CAPX)		\$120.0	\$0.0	\$0.0	\$0.0
dNet Working Capital (dNWC)		\$15.0	\$5.0	\$5.0	-\$25.0
Corporate Tax Rate (T)	25%				
Free Cash Flow (FCF)		?	?	?	?

\$135

\$30

From EBIT to FCF

- Calculate Free Cash Flow (FCF) from income statement as:

$$\text{FCF} = \text{EBIT} \times (1 - t_c) + \text{DEPR} - \text{CAPX} - \Delta\text{NWC}$$

- Start from earnings before interest and taxes (EBIT)
- Adjust for corporate taxes (t_c), depreciation (DEPR), capital expenditures (CAPX), and changes in net-working capital (ΔNWC)

Example: Calculating FCFs

Item (figures in \$'000)		Year 0	Year 1	Year 2	Year 3
Revenue (REV)		\$0.0	\$400.0	\$440.0	\$480.0
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dNet Working Capital (dNWC)		\$15.0	\$5.0	\$5.0	-\$25.0
Corporate Tax Rate (T)	25%				
Free Cash Flow (FCF)		-\$135.0	\$50.0	\$65.0	\$110.0
(= EBIT(1-T)+DEPR-CAPX-dNWC)					

Question

- Why are we going through all the trouble of setting up the 3-statement model, and all these calculations and adjustments, to forecast FCFs?
 - Why not just forecast FCFs directly?

Detour: Portfolio Returns

What is portfolio's return?

	<u>Asset</u>	<u>Return</u>	
A	\$100	10%	$\$10$ $?$ $\$60$
B	\$300	20%	

$\$70$
 $\frac{\$70}{\$100 + \$300} = \frac{\$70}{\$400}$
 $= 12.5\%$

$$S_A = \frac{\$100}{\$400} = 25\%$$

$$S_B = \frac{\$300}{\$400} = 75\%$$

$$(10\% \times 25\%) + (20\% \times 75\%) = 12.5\%$$

One calculation

	<u>Asset</u>	<u>Return</u>
A	\$100	10%
B	\$300	20%

- Asset A pays \$10 (10% of \$100)
- Asset B pays \$60 (20% of \$300)
- Total payoff is \$70 (\$10 + \$60)
- Portfolio value is \$400 (\$100 + \$300)
- Hence, portfolio return is 17.5% (\$70 / \$400)

Another calculation

	<u>Asset</u>	<u>Return</u>
A	\$100	10%
B	\$300	20%

- Asset A pays 10%
- Asset B pays 20%
- Weighted average of assets' returns is:
 - $10\% \times (\$100/\$400) + 20\% \times (\$300/\$400) = 17.5\%$

Portfolio return

- A portfolio's return, r_p , is the weighted average of the returns of its individual assets:

$$r_p = r_1 \frac{A_1}{A_1 + A_2} + r_2 \frac{A_2}{A_1 + A_2}$$

- Here, r_1 is asset 1's return and A_1 is value (same for asset 2)
 - Portfolio's total value is $A_1 + A_2$
 - Note, equation works both for realized and for expected (average) returns
- Equation also works for more than two assets (for any number of assets)

DCF/NPV in Practice: Discount Rate (WACC)

Discount Rate for DCF/NPV

- I want to value Amazon (AMZN)
 - Specifically, I want to value its operations
 - In other words, value the FCFs generated by its operations
 - This value is known as “Enterprise Value” (EV)
- As always, the correct discount rate is the expected return of an equally risky investment in the stock market
 - This would be an investment with the same systematic risk (beta) as Amazon’s FCFs and operations
- A natural starting point is the systematic risk (beta) of Amazon’s shares
 - However, this is systematic risk of Amazon’s equity (its shares), not its operations





Amazon.com Inc

✓ Added ▾

\$239.20 ↑ +42.94% (+71.86) 5Y

Jun 10, 12:41:23 PM UTC-4 · USD



Overview Earnings Financials Holdings New

📈 Amazon declined approximately 8% over the past month, pressured by a massive \$200 billion capital exp... ▾

Open	\$243.61	Volume	16.71M	Beta	1.45
High	\$250.43	P/E ratio	28.61	Shares outstanding	10.76B
Low	\$237.00	52-wk high	\$278.56	No. of employees	2M
Mkt. cap	2.58T	52-wk low	\$196.00		
Avg. vol.	44.07M	EPS	\$8.37		

Balance sheet view of company

Assets	Liabilities
Excess Cash (EC)	Debt (D)
Operating Assets (OA)	Equity (E)

- Company's total value is the value of its operating assets ("Enterprise Value")
 - Ignoring excess cash, for now
- Enterprise value is also the combined value of its debt and equity

Balance sheet view of company

Assets	Liabilities
Excess Cash (EC)	Debt (D)
Operating Assets (OA)	Equity (E)


$$\text{Value} = D + E$$

- Conceptually, a company's total value is like a portfolio consisting of the company's debt and its equity
 - For our discount rate, we need expected return of an investment in market that is as risky as our operating assets

Portfolio return of enterprise value (EV)

Assets	Liabilities
Excess Cash (EC)	Debt (D)
Operating Assets (OA)	Equity (E)

$$r_{EV} = r_D \frac{D}{D + E} + r_E \frac{E}{D + E}$$

- For now, ignore excess cash; pretend it is zero
- Enterprise value is not traded directly in the market
 - But debt and equity *are* traded; can get their expected returns in market
 - Note, E is typically much larger than D , so r_E is more important than r_D .

Return to equity (r_E)

- Return to equity (r_E) is the return earned by the company's shareholders
 - Both increases in stock price (P_t) and dividends (D_t)
 - Equity return is: $r_E = (P_1 + D_1)/P_0$
- Stock's expected return (r_E) can be determined from stock's beta (β_E) and CAPM
 - CAPM: $r_E = r_f + \beta_E \times r_{mf}$

Return to debt (r_D)

- Return to debt is the interest payments made to the lenders (creditors)
 - Can be determined from the yield to corporate bonds (with the same bond rating)
 - E.g., double-A rated bonds have a yield of 5.070%
- However, interest payments are tax deductible
 - In accounting, taxable earnings are calculated after deducting interest payments
- With 25% corporate taxes, a company can, for example, pay \$4 in interest to its creditors from just \$3 in earnings from its operations
 - Company gets the fourth dollar from the \$4 reduction in its taxable earnings and the resulting \$1 reduction in its corporate taxes (corporate taxes are 25%)
 - In general, it only costs company $(1-t_C)r_D$ to pay r_D in interest

Return to operating assets

Assets	Liabilities
Excess Cash (EC)	Debt (D)
Operating Assets (OA)	Equity (E)

Return to Operations Market return

$$(1-t_C)r_D \longrightarrow r_D$$

$$r_E \longrightarrow r_E$$

- In the market, expected return to debt and equity are r_D and r_E
- Expected return in the market with the same risk as company's operations is thus weighted average of $(1-t_C)r_D$ and r_E

Weighted Average Cost of Capital

$$WACC = (1 - t_C)r_D \frac{D}{D + E} + r_E \frac{E}{D + E}$$

- Interest payments are tax deductible ($t_C = 25\%$)
 - Paying \$1 in interest reduces taxable income by \$1 and reduces corporate taxes by t_C .
 - Company's net-of-tax cost of paying \$1 in interest to debt is only $(1-t_C)$
- Weighted Average Cost of Capital (WACC) is expected return on company's operations assets required to pay r_D in (expected) interest to debt and r_E in (expected) return to equity in the market
 - This is the discount rate for the DCF/NPV calculation

Excess cash and net debt

Assets	Liabilities		
Excess Cash (EC)	Debt (D)		Net Debt (D)
Operating Assets (OA)	Equity (E)		Equity (E)

- Formally, excess cash is cash that is not tied up in operations (can be paid out to investors)
 - In practice, determining excess cash from balance sheet (cash and marketable securities) can require some judgment
- Typically, the amount of excess cash is small, and it can be ignored
- Otherwise, subtract excess cash from debt, and do analysis with “net debt” (= debt – excess cash)